Mahesh Kumar Yarroju

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 3+ years of experience designing, building, and deploying AI and machine learning solutions across computer vision, LLM-based applications, and advanced data analytics. Proven expertise in Python, SQL, TensorFlow, Keras, and Power BI, along with modern generative AI frameworks such as RAG, LangChain, and LangGraph. Demonstrated success delivering end-to-end ML pipelines, improving model performance, and transforming complex datasets into scalable, production-ready solutions that drive business insights and decision-making.

TECHNICAL SKILLS

Programming	Python
Machine Learning	Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Deep Learning	Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Transfer Learning, Image Segmentation, Object Detection (YOLO)
Generative AI & NLP	Large Language Models (LLMs), Prompt Engineering, Retrieval Augmented Generation (RAG), Agentic AI, LangChain, LangGraph
Libraries & Frameworks	NumPy, Pandas, Scikit-learn, TensorFlow, Keras
Data Visualization	Power BI, Matplotlib, Seaborn
Databases	MySQL
Tools & Platforms	Jupyter Notebook, Google Colab, Git, N8N Workflow Automation
Operating Systems	Windows, Linux

PROFESSIONAL EXPERIENCE

Data Scientist
Asics Technologies Pvt Ltd (Client: TCS) – Hyderabad Feb 2023 – Present

- Designed, developed, and deployed machine learning and deep learning solutions for real-world business problems including computer vision, NLP, and sentiment analysis applications.
- Built and optimized advanced image processing systems including background removal and object detection using CNN architectures, YOLO models, and segmentation techniques, improving model accuracy and operational efficiency.
- Developed LLM-powered applications using fine-tuned GPT models, Retrieval-Augmented Generation (RAG) pipelines, and LangChain frameworks to enable intelligent content generation and adaptive learning workflows.
- Executed full end-to-end data science pipelines including data collection, preprocessing, feature engineering, model training, validation, and hyperparameter optimization to deliver high-performing ML models.
- Implemented real-time sentiment analysis systems for both audio and text datasets, enabling actionable insights for customer experience monitoring and CRM integration.
- Conducted large-scale customer behavior segmentation and automated feature selection across high-dimensional transactional datasets to support marketing strategy and data-driven business decisions.
- Developed interactive dashboards and analytical reports using Power BI, Matplotlib, and Seaborn to communicate insights and model results effectively to stakeholders.
- Collaborated with cross-functional engineering and product teams to integrate AI models into production environments while ensuring scalability, reliability, and maintainability.
- Maintained comprehensive documentation of experiments, workflows, and results while following best practices in version control using Git.

PROJECTS

Image Processing Project – Automated Background Removal

- Led an end-to-end deep learning initiative to develop an automated image background removal system using CNN-based segmentation models.
- Optimized model architecture and inference pipelines, improving precision and processing efficiency by approximately 20% and enabling smooth deployment in production environments.

LLM-Based Educational Platform (Generative AI + RAG)

- Designed and implemented an intelligent learning assistant powered by LLMs to enhance student engagement, assessment quality, and interactive learning experiences.
- Fine-tuned GPT-based models to enable dynamic content generation and adaptive evaluation, supporting personalized learning workflows.

Object Detection Framework for Drone Imagery

- Developed a Foreign Object Debris (FOD) detection system using drone imagery by implementing YOLO object detection and SAM-based segmentation models.
- Enabled real-time object detection and localization, improving detection accuracy by 25%.
- Built and trained deep learning pipelines using TensorFlow and Keras, including data preprocessing, model tuning, and performance optimization across diverse datasets.

Real-Time Audio Sentiment Analysis – Automobile Industry

- Built a real-time sentiment analysis platform to classify inbound and outbound customer calls as positive, negative, or neutral across English and Hindi languages.
- Integrated sentiment analytics into CRM systems, enabling improved feedback analysis, customer satisfaction monitoring, and service quality evaluation.

Feature Selection & Customer Behavior Segmentation

- Processed large-scale transactional datasets containing 1,000+ features and designed automated feature selection pipelines using Python.
- Applied clustering algorithms to segment customers and provide actionable insights to marketing teams, improving campaign targeting and strategic decision-making.

EDUCATION

B.Tech – Computer Science & Engineering
Chalapathi Institute of Technology – Guntur — May 2019
GPA: 73%

Intermediate – MPC
Santhinikethan Junior College — Apr 2015
GPA: 94.4%

SSC
S.R.K.H High School — Apr 2013
GPA: 9.3

CERTIFICATIONS

Data Science Certification
Innomatics Research Labs